

Higher Order Corrections:

- Deduce conditions for renormalizability.
- Look at scalar field theory in d -dim.

$$\mathcal{L} = -\frac{1}{2} Z_\phi \cdot \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} Z_m m^2 \phi^2 - \sum_{n=3}^{\infty} \frac{1}{n!} Z_n g_n \phi^n$$

- Consider a Feynman diagram:

E : External lines $\sim \langle \Omega | \phi_1 \dots \phi_E | \Omega \rangle$

I : Internal lines $\sim g_n^I$?

L : Closed loops

V_n : Vertices that connect n -lines

- Each closed loop gives a factor of $d L_i$ and each internal propagator gives a factor of $1/(p^2 + m^2)$

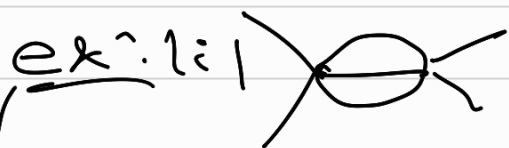
* Superficial degree of divergence:

$$\equiv \left(\prod_{i=1}^n \int \frac{d^d l_i}{(2\pi)^d} \right) \left(\frac{l_i^a}{l_i^b} \right)$$

$$\Rightarrow \boxed{SDOF = a - b} \quad \text{or}$$

$$\boxed{D = dL - 2I}$$

* Diagram is divergent if $\underline{D \geq 0}$
 $dL \geq 2I$



$$L = 2$$

$$d \geq 2$$

$$I = 2$$

$$L = 1$$

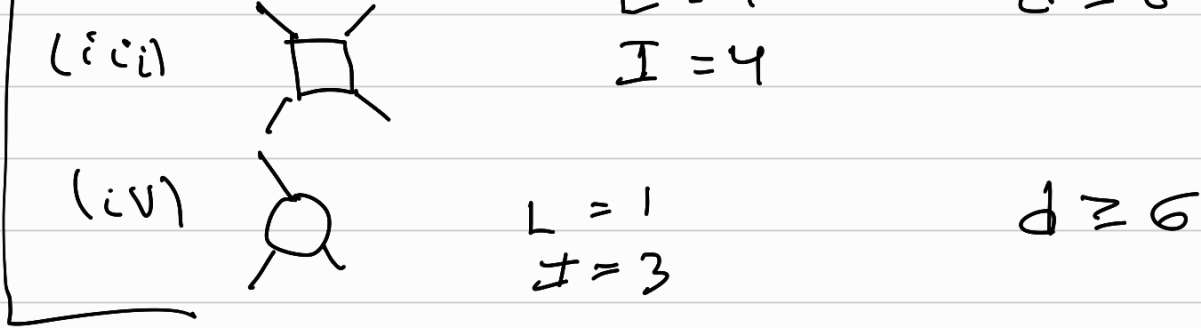
$$d \geq 4$$

$$I = 2$$

$$L = 1$$

$$d \geq 8$$





Vertex factor: $-iZ_E g_E$

Original tree diagram has mass dim $[g_E]$ and so

$$[\text{diagram}] = [g_E]$$

where

$$[\text{diagram}] = dL - 2I + \underbrace{\sum_{n=3}^{\infty} V_n [g_n]}_D$$

Thus,

$$D = [g_E] = \sum_{n=3}^{\infty} V_n [g_n]$$

Recall divergences occur if $D \geq 0$. If $[g_n] < 0$ for any n then D is strictly ≥ 0 and diverges.

For $g_n \phi^n$: $[g_n] = d - \frac{1}{2}n(d-2)$

$$[g_n] < 0 \Rightarrow n > 2d/(d-2)$$

(i) $d=4$: $n > \frac{8}{2} \Rightarrow n > 4$

so we are limited to theories of

order ϕ^4 $n > \frac{12}{4} = 3$

(ii) $d=6$: $n > \frac{12}{4} = 3$

so we are restricted to $\phi(\phi^3)$

* A theory is nonrenormalizable if the coeff. of any term in the Lagrangian

has negative mass dim.

why D?

• L: a loop yields $\int \frac{d^d k_i}{(2\pi)^d} = d$
so if there are L loops we get

$$Ld$$

• I: each internal line is a prop
 $1/(p^2 + m^2)$

and gives a $\int \frac{1}{p^2} = 2$

$$\therefore = 2I$$

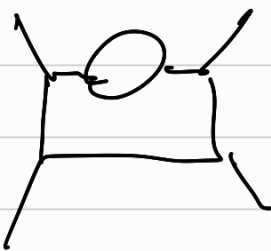
$$\Rightarrow D = Ld - 2I$$

ex: (a) V_4



$$E = 4$$

(b)



+



• The bubble diagram at 2-loop diverges

$$D = 2L - 2I$$

$$= 2(2) - 2(7) = -10?$$

* Divergent sub-diagrams are diagrams that, considered in isolation, have $D \geq 0$

Ex. 1 | Consider a massless Dirac field in (a) d-dim.

$$\mathcal{L} = i\bar{\psi} \gamma^\mu \partial_\mu \psi$$

$$\bar{\psi} = \psi^\dagger \gamma^0$$

$$-d + 1 + 2[4] = 0$$

$$[4] = (d-1)/2$$

(b) Consider $g_n (\psi \psi^\dagger)^n$, $n \geq 2$
 $[g_n] + (n/2)(d-1) - d = 0$
 $[g_n] = d - n(d-1)$

(c) $g_{m,n} \psi^m (\psi \psi^\dagger)^n$
 $[g_{m,n}] + \frac{m}{2}(d-2) + (n/2)(d-1) - d = 0$
 $[g_{m,n}] = d - \left(\frac{m}{2}(d-2) + n(d-1) \right)$

(d) For $d=4$
 (i) $g_n (\bar{\psi} \psi)^n$, $n \geq 2$
 $[g_n] = 4 - n(3) \geq 0$
 $n(3) \leq 4 \Rightarrow n \leq 4/3$
 So for a renormalizable theory
 $n=1$

which is $g_1 (\bar{\psi} \psi)$
 which is precisely the mass term
 $[g_1] = 1 \Rightarrow g_1 = m$
 (ii) $g_{m,n} \psi^m (\bar{\psi} \psi)^n$

$[g_{m,n}] = 4 - [m + n(3)] \geq 0$
 $m + 3n \leq 4$
 $n=1 : \begin{cases} m=0 \\ m=1 \end{cases}$

Highest order
 $g_{1,1} \psi (\bar{\psi} \psi)$